Keywords: diffusion models • random matrix theory • generative models • statistical learning

The Seed Knows the Model

Random Matrix Theory Reveals Why Diffusion Models Trained on Different Data Produce the Same Images

By Binxu Wang, Jacob A. Zavatone-Veth & Cengiz Pehlevan (Harvard Kempner Institute; Harvard Society of Fellows; Harvard SEAS)

arXiv:2602.02908 • ICML 2026 Oral / Honorable Mention

Train two diffusion models on non-overlapping halves of ImageNet. Give both the same noise seed. Despite never sharing a single training image, the models produce strikingly similar outputs. This reproducible observation has prompted concern about memorization and dataset leakage —but the true cause is far more mundane and, paradoxically, more fundamental: shared Gaussian statistics across any large natural image dataset already constrain most of what a finite-data denoiser can generate. A random matrix theory framework makes this precise, predicting cross-split consistency from first principles.

AT A GLANCE

Problem: Diffusion models trained on disjoint data subsets generate nearly identical images from the same noise seed, raising memorization and reproducibility concerns.

Why it matters: Understanding the source of this consistency is critical for auditing generative models for data privacy, copyright, and reproducibility.

Method: Random matrix theory analysis of the linear diffusion model; exact formulas for the expected denoiser and its cross-split variance.

Result: Consistency traces to shared Gaussian statistics across splits; the effective noise level is renormalized as $\sigma^2 \to \kappa(\sigma^2)$ via a self-consistent RMT equation.

Evidence: RMT predictions match empirical cross-split cosine similarity within 2–4% across all tested dataset sizes and aspect ratios.

The Big Picture

Diffusion models generate images by iteratively denoising a noise vector $x_T \sim \mathcal{N}(0, I)$. Because the entire generation process is deterministic given the initial seed x_T, two models starting from the same seed produce the same image if and only if they have learned the same score function. The observation that models trained on *different* data nonetheless produce similar images therefore implies that the learned score functions agree — which in turn implies that the differences in training data have largely not been captured.

Prior work attributed this phenomenon to memorization (the models share training examples through implicit data augmentation) or to architectural inductive biases (convolutional symmetries, attention patterns). This paper shows neither explanation is necessary: the consistency is predicted by the population-level Gaussian statistics of the data, which are approximately identical across any two large-enough random subsets of natural images.

Why Existing Approaches Fall Short

Memorization-based accounts predict that cross-split consistency should decrease as the two data splits become more diverse. However, the consistency persists even when the splits are drawn from entirely different semantic categories of ImageNet, falsifying this explanation.

Architecture-based accounts predict that consistency should vanish for architectures trained without weight sharing (e.g., plain MLPs). Empirically, consistency is observed across all tested architectures, inconsistent with this account.

Statistical privacy audits (membership inference, differential privacy bounds) focus on individual example memorization rather than population-level structural similarities. None of these frameworks could predict the direction or magnitude of cross-split agreement.

Core Mechanism

The consistency mechanism operates through the spectral structure of the data covariance. In the linear diffusion model, the optimal denoiser at noise level σ is the Wiener filter:

$$s_\sigma(x_\sigma) = -(\Sigma + \sigma^2 I)^{-1} x_\sigma$$

where Σ is the data covariance. Two models trained on different splits both estimate Σ from their respective finite datasets. Their estimates $\hat{\Sigma}^{(1)}$ and $\hat{\Sigma}^{(2)}$ are noisy, but their *mean* over random splits converges to the *same* population covariance Σ. The shared population covariance is the source of cross-split consistency: the two denoisers agree in expectation and differ only in their finite-sample fluctuations, which vanish at rate $O(1/\sqrt{N})$.

Technical Formulation

In the proportional limit $N, d \to \infty$ with $\gamma = d/N$ fixed, the sample covariance resolvent $(\hat{\Sigma} + \sigma^2 I)^{-1}$ converges to a deterministic function of the population covariance Σ via the self-consistent equation:

$$\kappa(\sigma^2) = \sigma^2 + \gamma \int \frac{\lambda}{\lambda + \kappa(\sigma^2)} \, d\mu(\lambda)$$

where μ is the spectral measure of Σ. This $\kappa(\sigma^2) > \sigma^2$ is the *effective noise level*: finite-data denoisers experience more noise than they receive, explaining the systematic bias toward the dataset mean.

The expected cross-split agreement of generated samples is:

$$\mathbb{E}[\langle x^{(1)}, x^{(2)} \rangle] = \mathrm{tr}\left[(\Sigma + \kappa I)^{-2}\, \Sigma^2\right] / d$$

Cross-split variance decomposes into three additive terms:

$$\mathrm{Var}[\hat{s}_\sigma(x)] = V_{\text{aniso}} + V_{\text{inhom}} + V_{\text{size}}$$

where V_{aniso} captures eigenmode anisotropy, V_{inhom} captures per-input inhomogeneity, and $V_{\text{size}} = O(1/N)$ is the global $1/N$ scaling.

Learning or Inference Procedure

1. Draw two independent data splits of size N each from $p(x)$.
2. Train a linear diffusion model (Wiener denoiser) on each split, obtaining score functions $\hat{s}_\sigma^{(1)}$ and $\hat{s}_\sigma^{(2)}$.
3. Generate images $x^{(1)}$ and $x^{(2)}$ from the same noise seed x_T using Euler–Maruyama sampling.
4. Measure cross-split cosine similarity $\langle x^{(1)}, x^{(2)} \rangle / (\|x^{(1)}\| \|x^{(2)}\|)$.
5. Compare to the RMT prediction derived from the population covariance Σ and $\kappa(\sigma^2)$.

The RMT prediction requires only Σ and $\gamma = d/N$ — neither the specific split realizations nor any model training is needed.

What the Guarantee Says

The main theorem establishes: as $N, d \to \infty$ with $\gamma = d/N$ and signal-to-noise ratio bounded away from 0 and ∞:

(1) *Expectation:* $\mathbb{E}[\hat{s}_\sigma(x)]$ converges to $-(\Sigma_{\text{eff}} + \kappa I)^{-1} x$ where Σ_{eff} is a deterministic function of the population covariance.

(2) *Fluctuations:* $\mathrm{Var}[\hat{s}_\sigma(x)] = O(1/N)$, so cross-split agreement is $1 - O(1/N)$ for fixed d/N.

(3) *Bias:* Generated samples are systematically pulled toward the dataset mean along low-variance principal components, with magnitude $O(\gamma)$ for fixed spectral gap.

Experimental Findings

Setting	Measured	RMT pred.
Linear, $N = 500$	0.47	0.46
Linear, $N = 2000$	0.23	0.22
$\gamma = 2.0$	0.61	0.60
ImageNet PCA approx.	0.38	0.36

All values are cross-split cosine similarity. RMT predictions match observations within 2–4% across all settings.

The framework also correctly predicts *directional* bias: generated samples shift toward the dataset mean along the eigenvectors with smallest eigenvalues, which the Wiener filter systematically undershoots.

Ablations and Interpretation

Dataset size: Cross-split consistency decreases as $O(1/\sqrt{N})$, confirming the $1/N$ fluctuation bound. Practitioners can use this formula to determine the minimum dataset size needed to keep consistency below an acceptable threshold.

Aspect ratio γ: Higher γ (more parameters per sample) increases cross-split consistency. Modern large-latent-space models effectively operate at high γ, explaining why they show stronger consistency than smaller models.

Nonlinear diffusion: Experiments on DDPM with U-Net score models show qualitatively consistent results: cross-split similarity decreases with N and increases with γ, suggesting the linear theory captures the dominant statistical effect even in nonlinear models.

Semantic consistency: Cross-split consistency in semantic embedding space (CLIP) is 2–3× larger than in pixel space, suggesting the linear mechanism dominates more strongly at the semantic level where principal components carry more variance.

Reference

Binxu Wang, Jacob A. Zavatone-Veth, Cengiz Pehlevan. **A Random Matrix Theory Perspective on the Consistency of Diffusion Models**. arXiv:2602.02908, February 2026. ICML 2026 Oral / Outstanding Paper Honorable Mention. https://arxiv.org/abs/2602.02908